
DS & AI - Seeing through the hype [edition 2026/2027]

Data Quality (DQ)

Lab 2: Detection and Cleaning - Exam preparation

Introduction

Goal The goal of this lab is to practice with concepts and techniques for detection of data quality problems as well as cleaning or mitigating them. Most exercises are on-paper exercises, so mostly no programming is needed. The style of questions is on purpose very similar to the exam questions, so this lab is at the same time exam preparation

Overview You will do this lab in the same groups of three (self-enrollment via Canvas) as with DQ Lab 1. The exercises have an indicated difficulty level: easy (for a sufficient number of 45 points), regular (for up to 60 points), and advanced (for up to 70 points).

Assessment There are two labs for the DQ part. Lab 1 (which you already did) amounts to 30 points and Lab 2 (this one) amounts to 70 points. The sum of the points of both labs determines the DQ lab grade. This grade has weight 20% for the final grade of the DQ part.

$$DQ \text{ Lab Grade} = \frac{1}{10}(Points \text{ lab } 1 + Points \text{ lab } 2) \quad (1)$$

$$Final \text{ DQ grade} = \frac{1}{5}(DQ \text{ Lab Grade}) + \frac{4}{5}(DQ \text{ Test Grade}) \quad (2)$$

There is a minimum requirement of a 5.5 on the DQ lab grade. Since the lab grade is based on both labs, the one lab could compensate for a bad performance on the other lab. There is a repair, but it is for both labs together and you can only achieve a maximum of 60 points for the repair (i.e., a DQ lab grade of 6.0).

NB: In contrast to Lab 1, you are expected to make all exercises.

Deadline, deliverables and mode of delivery There is a spreadsheet file in which you can fill in your answers: `DSAI-DQ-Lab-2-Answer-Sheet.xlsx`. Blue cells are for multiple choice questions and contain a dropdown list for the possible answers. Green cells are for fill-in questions where you only need to fill in something short. Yellow cells are larger to give you space, because these are open questions. You need to submit your answers as a filled in answer sheet on Canvas. The intention is that you can finish the lab during the scheduled 4-hour lab session. For reasons of flexibility and resiliency to exceptions, the deadline is a week later; check the Canvas assignment for the exact submission deadline. This is a very strict deadline, because of the following:

The morning after the deadline, the answers to the lab exercises will be published. The reason for this is that you can use these exercises for studying for the exam.

Example database

Some of the questions refer to an example case concerning a flight reservation system. An excerpt of the data of this example database is shown below. It contains two tables: “Reservation” contains the reserved tickets and “Flights” contains information about flights.

Reservation

rid	cid	fnm	lnm	fid	day	time	bl	fcode	fcity	tcode	tcity	plane
1	1	John	Lawson	KL123	5/9/22	9:00	B	AMS	Amsterdam	VIE	Vienna	A380
1	2	Mary	Lawson	KL123	5/9/22	9:00	L	AMS	Amsterdam	VIE	Vienna	A300
2	3	Peter	Blanks	KL123	5/9/22	9:00		AMS	Amsterdam	VIE	Vienna	A380
3	4	Floor	Dessing	KL286	6/9/22	8:00	B	AMS	Amsterdam	HND	Tokyo	A330
4	1	Jon	Lawson	LH214	12/9/22	7:00		FMO	Münster	MUC	Munich	B747
5	2	Mary	Lawson	KL123	12/9/22	9:00	L	AMS	Amsterdam	VIE	Wenen	A380

Flights

fid	day	time	fcode	fcity	fcountry	tcode	tcity	tcountry	km
KL123	Mon	9:00	AMS	Amsterdam	NL	VIE	Vienna	Austria	1000
KL286	Tue	8:00	AMS	Amsterdam	NL	HND	Tokyo	Japan	5000

Description of attributes

rid	Reservation identifier. It is possible to reserve more than one ticket per reservation
cid	Customer identifier
fnm	Firstname of the customer
lnm	Lastname of the customer
fid	Flight ID. This is a unique code for a weekly recurring flight, e.g., KL123 is the code for the flight from Amsterdam to Vienna that flies every week on Mondays departing 9:00.
day	In the Reservation table, the exact date of the ticket. In the Flights table, the day of the week of the flight.
time	The departure time of the flight.
bl	Either ‘B’ for Business or ‘L’ for Leisure (or missing).
fcode	Code of the departure airport (i.e., ‘from code’)
fcity	City of the departure airport.
fcountry	Country of the departure airport.
tcode	Code of the destination airport (i.e., ‘to code’)
tcity	City of the destination airport.
tcountry	Country of the destination airport.
plane	The type of airplane used for the flight.
km	The distance flown.

Exercises

Exercise 1 (Constraints)

27 pts

These questions are about the formulation of constraints, the detection of constraint violations, and cleaning of data that violates constraints.

(a) [8pts] (Easy)

Which of the following constraints should hold for the Reservation table of the flight reservation system? The first four are functional dependencies, the last two are inclusion dependencies. *NB: Do not look at the data (that is the next question), only base your answer on the description of the attributes and your knowledge of how flying and airports work.*

- ☐ $\text{cid} \rightarrow \text{fnm}, \text{lnm}$
- ☐ $\text{fid} \rightarrow \text{day}, \text{time}$
- ☐ $\text{fid} \rightarrow \text{fcode}, \text{fcity}, \text{tcode}, \text{tcity}$
- ☐ $\text{rid} \rightarrow \text{cid}$
- ☐ $\text{Reservation}[\text{fid}] \subseteq \text{Flights}[\text{fid}]$
- ☐ $\text{Flights}[\text{fid}] \subseteq \text{Reservation}[\text{fid}]$

(b) [6pts] (Easy)

Which of the following constraints hold for the Reservation table of the flight reservation system? *NB: Now look at the data.*

- ☐ $\text{cid} \rightarrow \text{fnm}, \text{lnm}$
- ☐ $\text{rid} \rightarrow \text{fcode}, \text{tcode}, \text{plane}$
- ☐ $\text{fcode} \rightarrow \text{fcity}$
- ☐ $\text{tcode} \rightarrow \text{tcity}$

(c) [2pts] (Regular)

Which is the correct formal notation for the constraint $\text{Flights} : \text{fid} \rightarrow \text{fcode}, \text{tcode}$?

- ☐ $\forall a_1 \forall a_2 \forall b \forall c \forall d \forall e \forall f \forall g \forall h \forall i \forall j (\text{Flights}(a_1, b, c, d, e, f, g, h, i, j), \text{Flights}(a_2, b, c, d, e, f, g, h, i, j) \Rightarrow a_1 = a_2)$
- ☐ $\forall a \forall b \forall c \forall d_1 \forall d_2 \forall e \forall f \forall g_1 \forall g_2 \forall h \forall i \forall j (\text{Flights}(a, b, c, d_1, e, f, g_1, h, i, j), \text{Flights}(a, b, c, d_2, e, f, g_2, h, i, j) \Rightarrow d_1 = d_2 \wedge g_1 = g_2)$
- ☐ $\forall a \forall b_1 \forall b_2 \forall c_1 \forall c_2 \forall d_1 \forall d_2 \forall e_1 \forall e_2 \forall f_1 \forall f_2 \forall g_1 \forall g_2 \forall h_1 \forall h_2 \forall i_1 \forall i_2 \forall j_1 \forall j_2 (\text{Flights}(a, b_1, c_1, d_1, e_1, f_1, g_1, h_1, i_1, j_1), \text{Flights}(a, b_2, c_2, d_2, e_2, f_2, g_2, h_2, i_2, j_2) \Rightarrow d_1 = d_2 \wedge g_1 = g_2)$

(d) [3pts] (Easy)

A conditional functional dependency (CFD) is a pair $(X \rightarrow Y, T_p)$. What is the correct technical term for T_p ?

- ☐ Table pattern

- ☐ Tableau pattern
- ☐ Pattern table
- ☐ Pattern Tableau

(e) [4pts] <Easy>

_____ If a constraint is violated in a database, does this also always indicate what data record is wrong?

(f) [2pts] <Regular>

Which of the names will appear in the result of the query below under the Consistent Query Answering approach given that the constraint $cid \rightarrow fnm, lnm$ should hold?

```
SELECT DISTINCT fnm, lnm FROM Reservation
```

(g) [2pts] <Advanced>

Which of the names will appear in the result of the query below under the Consistent Query Answering approach given that all constraints of exercise 1(b) should hold?

```
SELECT fnm, lnm FROM Reservation WHERE tcode='VIE'
```

Exercise 2 (Answer Set Programming)

5 pts

For the exercise on Answer Set Programming, `DSAI-DQ-GenerateRepairsForFD.ipynb` is a Python notebook containing a hands-on step-by-step demonstration how Answer Set Programming can be used to automatically determine a most likely repair (or repairs) for a table that contains violations of a functional dependency constraint.

Instructions: Download the file from Canvas. Follow the steps in the notebook, carry out the exercises, and answer the questions below based on the results you got. The exercises are code completion exercises, where you are asked to fill in a few mission lines of code.

(a) [1pts] <Regular>

The code in the section on “Functional dependency” determines all violations of the functional dependency in the example data. Which violations are there?

(b) [1pts] <Regular>

The code in the section on “Generating all possible repairs” determines with function ‘`apply_repairs`’ all possible repaired tables that solve the violations. How many possible repaired tables does it generate (unfiltered)?

(c) [3pts] <Advanced>

The code in the section on “Generating all possible repairs” determines in the end the repaired tables that were repaired with the least cost. If all is well, it shows that there are 2 possible repaired tables that were obtained with repairs with minimal cost 2. This is strange: there were 3 violations, so how can all 3 violations be solved

with only 2 update repairs? Examine the verbose output of the code in this section to find out which single update repair apparently solved two violations in one go.

Exercise 3 (Duplicates)

13 pts

(a) [3pts] <Easy>

Calculate the Jaccard coefficient for strings “Jon” and “John” where we define the tokens as the individual characters. The formula for the Jaccard coefficient is

$$Jaccard(P, Q) = \frac{|P \cap Q|}{|P \cup Q|}$$

(b) [3pts] <Easy>

Calculate the Normalized Levenshtein for strings “Jon” and “John” where we define the tokens as the individual characters. The formula for Normalized Levenshtein is

$$N_{Levenshtein}(A, B) = 1 - \frac{L(A, B)}{\max(\text{length}(A), \text{length}(B))}$$

where $L(A, B)$ is the number of manipulations to turn A into B .

(c) [2pts] <Regular>

We conduct a duplicate detection on the names of customers in the ‘Reservation’ table. We use as lower threshold $T_\lambda = 0.4$ and upper threshold $T_\mu = 0.9$. For brevity, let’s define the following shorthands for our names: A =“Jon Lawson”, B =“John Lawson”, C =“Mary Lawson”, D =“Peter Blanks”, E =“Floor Blanks”. The duplicate detection tool produced the following similarity matching scores.

	A	B	C	D	E
A	1.00	0.95	0.45	0.20	0.25
B		1.00	0.50	0.20	0.25
C			1.00	0.25	0.20
D				1.00	0.45
E					1.00

Give all possible partitionings given the thresholds.

(d) [3pts] <Advanced>

Take one of your possible partitionings and calculate its probability. Explain your answer by giving the full calculation.

(e) [2pts] <Advanced>

In the probabilistic data integration approach, we can store both possible partitionings in the database as uncertain data. Do they remain as two alternatives in the probabilistic database or is there a mechanism whereby the uncertainty can get resolved? Explain your answer.

Exercise 4 (Effect of Data Quality on Machine Learning and Data Imputation) 10 pts

Suppose we like to use our example database for training a model to predict whether a customer is a business traveler or someone who travels for leisure. The idea is that we could train a model on attributes like departure and destination locations, time of departure, airline, party size (i.e., number of persons in a reservation) to predict the “bl” attribute. The intended use of the model is for marketing purposes: offer relevant discounts and present relevant advertisements. Assume that we have much more data in our data than just the few rows shown.

(a) [4pts] <Easy>

_____ A customer could in reality be flying for business purposes the one time and for leisure purposes in the other. So, it could be that the same customer could have multiple reservations with mixed values for the ‘bl’ attribute. Is that a data quality problem?

(b) [4pts] <Easy>

Which of the following statements about dealing with data quality problems concerning this machine learning task, is true? (More than one answer possible)

- ☐ The ‘plane’ attribute is more trustworthy than the ‘lnm’ attribute.
- ☐ Deleting the rows with a missing value for the ‘bl’ attribute is called the “indicator method”.
- ☐ A low amount of errors in the data is acceptable.
- ☐ If the data would be 100% correct and accurate here, we could achieve 100% accuracy with our model.

(c) [2pts] <Regular>

_____ The ‘bl’ attribute has missing values. They could be missing completely at random (MCAR) or there could be a reason making it more likely to be missing under certain circumstances (i.e., MAR or MNAR). Is there a method to find out whether it is more likely to be MCAR or not?

Exercise 5 (Intro and Business perspective)

15 pts

These questions pertain to the first lecture with the motivation and general introduction to data quality and business perspective.

(a) [3pts] <Easy>

_____ Is it possible that the same data set would be of proved good quality in one context and be of bad quality in another? Explain your answer.

(b) [4pts] <Easy>

_____ Given a CSV-file with perfectly correct and accurate data in it. It

contains a sufficient amount of rows and columns and the contents are exactly what you need. Can it still be of bad quality? Explain your answer.

(c) [1pts] (Regular)

_____ Can “Security” be seen as a data quality problem? Explain your answer.

(d) [2pts] (Easy)

We have a training set of images $I_1, \dots, I_n$ containing birds each with a label y_i whether it is a canary bird or not. A Python program counts per image the number of yellow pixels p_i and determines the optimal threshold τ such that the number of images with that are indeed canary birds, is maximal. We also have a test set of bird images with labels with which we evaluate the accuracy of using this threshold for the number of yellow pixels. What kind of reasoning is this?

- ☐ Neural reasoning
- ☐ Neuro-symbolic reasoning
- ☐ Symbolic reasoning

(e) [2pts] (Advanced)

When I compared in the lecture symbolic reasoning with neural reasoning, I wrote “Interpretability” as “High (traceable)” for symbolic systems. Traceability is not the same as interpretability. Give an example case where a symbolic system is traceable but not interpretable.

(f) [3pts] (Regular)

In DQ lecture 1, I told the story of a PhD candidate of mine who focused on breast cancer diagnosis based on mammograms. A mammogram typically is a set of four pictures: an image is taken for each breast from the top and from the side. So, the idea was to develop a model that has 4 images as input and would predict ‘malignant’, ‘benign’ or ‘normal’. The PhD candidate found out that in her data, there was a data quality issue: 8% of the patients didn’t have four pictures. An easy fix would be to train the model on the cases they have exactly four pictures. They constitute 92% of the data, so still plenty data to successfully train a performant model on. Why did the PhD candidate not choose to do this?